

Got it 👍 — you want the **same 1-page theory**, but neatly in **points / subtopics format** (easy to write in exam).

I'll format each experiment properly 📌

✅ **Experiment 1 – Data Wrangling I**

♦ 1. Introduction to Data Wrangling

- * Data wrangling is the process of cleaning and transforming raw data.
- * It prepares data for analysis and machine learning.
- * Raw data is often incomplete, inconsistent, or unstructured.

♦ 2. Dataset Collection

- * Data is collected from open sources like Kaggle or UCI repository.
- * Example: Iris dataset with features like sepal length, petal width, etc.

♦ 3. Loading Data

- * Dataset is loaded into Python using Pandas.
- * ``read_csv()`` is commonly used.
- * Data is stored in DataFrame format.

♦ 4. Data Understanding

- * Use functions like:
 - * ``head()`` → view first rows
 - * ``describe()`` → summary statistics
 - * ``info()`` → data types and null values
- * Helps understand structure and features.

♦ 5. Handling Missing Values

- * Missing values represented as NaN or None.
- * Techniques:
 - * Remove rows/columns
 - * Replace with mean/median/mode

♦ 6. Data Formatting

- * Convert incorrect data types into proper format.

* Example: string to integer or date format.

♦ 7. Data Normalization

* Scaling values to a common range.

* Improves model performance.

♦ 8. Encoding Categorical Data

* Convert categorical values into numeric form.

* Example: Label Encoding.

Experiment 2 – Data Wrangling II

♦ 1. Introduction

* Focus on advanced preprocessing techniques.

* Improves data quality and reliability.

♦ 2. Missing Value Detection

* Functions used:

* ``isnull()`` → detects missing values

* ``notnull()`` → detects valid values

♦ 3. Handling Missing Values

* Methods:

* ``dropna()`` → remove missing data

* ``fillna()`` → replace with mean/median

* ``replace()`` → substitute values

♦ 4. Outlier Detection

* Outliers are extreme values.

* Can distort analysis.

* Identified using statistical methods or visualization.

♦ 5. Handling Outliers

* Remove extreme values

- * Replace with mean/median
- * Use transformation techniques

♦ 6. Data Transformation

- * Used to:
 - * Normalize data
 - * Reduce skewness
 - * Convert non-linear to linear relationships

Experiment 3 – Descriptive Statistics

♦ 1. Introduction

- * Used to summarize and analyze dataset.
- * Helps understand data distribution.

♦ 2. Measures of Central Tendency

- * **Mean** → Average value
- * **Median** → Middle value
- * **Mode** → Most frequent value

♦ 3. Measures of Dispersion

- * **Range** → Max - Min
- * **Standard Deviation** → Spread of data
- * **Variance** → Measure of variability

♦ 4. Importance

- * Helps in decision making.
- * Provides quick summary of dataset.

Experiment 4 – Linear Regression

♦ 1. Introduction

- * Supervised learning algorithm.

* Used to predict continuous values.

♦ 2. Concept of Linear Regression

* Relationship between dependent and independent variables.

* Represented as:

$$* Y = mX + b$$

♦ 3. Types

* Simple Linear Regression → one variable

* Multiple Linear Regression → multiple variables

♦ 4. Least Squares Method

* Minimizes error between actual and predicted values.

♦ 5. Model Evaluation

* **MSE (Mean Squared Error)**

* **RMSE (Root Mean Squared Error)**

* **R² Score**

♦ 6. Training and Testing Data

* Dataset divided into training and testing sets.

Experiment 5 – Logistic Regression

♦ 1. Introduction

* Used for classification problems.

* Output is categorical (0 or 1).

♦ 2. Logistic Regression Concept

* Predicts probability of an event.

* Uses logit function.

♦ 3. Sigmoid Function

* Converts values into range (0 to 1).

* Used for classification.

♦ 4. Types

- * Binary → 2 classes
- * Multinomial → multiple classes
- * Ordinal → ordered classes

♦ 5. Linear vs Logistic Regression

- * Linear → continuous output
- * Logistic → categorical output

♦ 6. Confusion Matrix

- * Used to evaluate classification:
 - * TP, TN, FP, FN

♦ 7. Evaluation Metrics

- * Accuracy
- * Precision
- * Recall
- * Error rate

Experiment 6 – Naïve Bayes

♦ 1. Introduction

- * Probabilistic classification algorithm.
- * Based on Bayes Theorem.

♦ 2. Bayes Theorem

- * Calculates probability using prior knowledge.

♦ 3. Assumption

- * All features are independent.

♦ 4. Types of Probabilities

- * Prior probability
- * Conditional probability

♦ 5. Working

- * Calculate probability for each class.
- * Choose class with highest probability.

♦ 6. Evaluation

- * Confusion matrix and metrics.

Experiment 7 – Text Analytics (TF-IDF)

♦ 1. Introduction to Text Mining

- * Extract meaningful information from text data.

♦ 2. NLP Operations

- * Tokenization
- * Stopword removal
- * Stemming
- * Lemmatization
- * POS Tagging

♦ 3. Bag of Words

- * Represents text as frequency of words.

♦ 4. TF-IDF

- * TF → term frequency
- * IDF → importance of word
- * Helps in feature extraction.

Experiment 8 – Data Visualization I

♦ 1. Introduction

- * Visualization helps understand patterns in data.

♦ 2. Seaborn Library

- * Used for statistical plots.
- * Built on matplotlib.

♦ 3. Types of Plots

- * Distribution plots (histogram, distplot)
- * Categorical plots (bar, box, violin)
- * Joint plots

♦ 4. Dataset

- * Titanic dataset used for analysis.

♦ 5. Importance

- * Helps identify trends and patterns.

Experiment 9 – Data Visualization II

♦ 1. Exploratory Data Analysis

- * Understand dataset before modeling.

♦ 2. Univariate Analysis

- * Study one variable.
- * Plots:

- * Histogram
- * Count plot
- * Pie chart

♦ 3. Bivariate Analysis

- * Study relationship between two variables.

♦ 4. Box Plot

- * Shows distribution and outliers.

Experiment 10 – Data Visualization III

♦ 1. Introduction to Pandas

- * Used for data analysis.

♦ 2. Dataset Handling

- * Load dataset using `read\_csv()`

♦ 3. Data Exploration

- * Functions:

- * `head()`
 - * `columns`
 - * `shape`

♦ 4. Visualization

- * Histogram
- * Boxplot

♦ 5. Outlier Detection

- * Identify abnormal values.

Experiment 11 & 12 – MapReduce

♦ 1. Introduction

- * Framework for processing big data.

♦ 2. MapReduce Concept

- * Map → convert data into key-value pairs
- * Reduce → aggregate results

♦ 3. Stages

- * Map stage
- * Shuffle stage

- * Reduce stage

♦ 4. Advantages

- * Scalable
- * Parallel processing
- * Handles large datasets

♦ 5. Applications

- * Weather analysis
- * Log processing

✅ This format is:

- * Easy to **write in exam**
- * Covers **main theory properly**
- * Not too long, not too short

If you want next:

👉 I can also give **ultra-short (5–6 mark answers)** OR

👉 Convert this into **perfect journal write-up format (with headings + paragraphs)**